Simulation Registry

All simulations share the same two-step loading protocol: - **Step-0**: inflate internal gas cavities to pressure `P` (volume-dependent fluid cavity) - **Step-1**: uniaxial compression ~75% in the **x-direction** (displacement = −15 on a 20-unit domain), recording S11, S12, S22 and x-face reaction forces RF2

The key variables across series are **mesh geometry**, **mesh type** (hexagonal vs. random), **boundary periodicity**, **Young's modulus E**, and **pre-inflation pressure P**. Material: E (MPa), ν = 0.3, plane-stress (CPS3/CPS4), t = 1, gas: N_2 at 273 K, P_0 = 0.101325 MPa.

Series Overview

Series	Sim numbers	Mesh files	Mesh type	Periodic	E (MPa)	P (MPa)	# Sims	Ran?
SIM_000	000	A004	Hexagonal	None	20	0.01	1	✅
SIM_001	001–003	A001–A003	Hexagonal	None	20	0.01	3	✅
SIM_100	100	A00A (test)	Hexagonal	None	20	0.01	1	✅
SIM_1000	1000–1039	A010–A013	Hexagonal	None	20	0–0.64 (9 lvls)	~36	❌
SIM_1100	1100–1139	A020–A023	Hexagonal	None	20	0–0.64 (9 lvls)	~36	❌
SIM_1200	1200–1208	A020	Hexagonal	None	20	0–0.64 (9 lvls)	9	❌
SIM_1210	1210–1218	A020	Hexagonal	None	20	0–0.64 (9 lvls)	9	❌
SIM_1300	1300–1308	A030	Hexagonal	lr	20	0–0.64 (9 lvls)	9	❌
SIM_1310	1310–1318	A030	Hexagonal	lr	20	0–0.64 (9 lvls)	9	❌
SIM_1320	1320–1328	A031	Hexagonal	lr	20	0–0.64 (9 lvls)	9	❌
SIM_2000	2000–2008	A2000	Hexagonal	both	20	0–0.64 (9 lvls)	9	❌
SIM_3000			Hexagonal	both			90	❌*

Series	Sim numbers	Mesh files	Mesh type	Periodic	E (MPa)	P (MPa)	# Sims	Ran?
	3000–3228	A3000, A3100, A3200			20, 10, 5	0–1.28 (10 lvls)		
SIM_4000	4000–4002	A3000	Hexagonal	both	20	0.64	3	❌
SIM_5000	5010–5428	R6000–R6004	**Random**	both	20, 10, 5	0–0.64 (9 lvls)	135	✅
SIM_6000	6100–6428	R6010–R6014	**Random**	both	20, 10, 5	0–0.64 (9 lvls)	135	✅
SIM_9000	9000	A020	Hexagonal	None	20	0	1	✅
SIM_9001	9001	TEST_P	Hexagonal	lr	20	0	1	✅
SIM_9002	9002	A031	Hexagonal	lr	20	0.02	1	✅
SIM_9003	9003	A00B (rect)	Hexagonal	ul	20	0.02	1	✅
SIM_9004	9004	A031	Hexagonal	both	20	0.02	1	✅
SIM_9005	9005	A991	Hexagonal	both	20	0.02	1	✅
SIM_9998	9998	A9998	Hexagonal	both	20	0.005	1	✅

*SIM_3000: creator scripts exist (`SIM_3000_inp_creator.py`, `SIM_3000_inp_creator_OnLY_9.py`) but no simulation folders found in `E001_Simulations/`.

Periodicity legend: `None` = free boundaries, `lr` = left-right periodic, `both` = fully periodic (left-right + top-bottom).

Detailed Series Descriptions

SIM_000 — Single baseline, no periodicity

- **Mesh**: `A004_hexagonal_mesh.hex2`, 1 realization, 20×20 scale
- **Loading**: P = 0.01 MPa inflation → x-compression −15
- **Purpose**: initial proof-of-concept / single-shot test

SIM_001 — Mesh realization variability (no periodicity)

- **Meshes**: `A001`, `A002`, `A003` (3 realizations of the same hexagonal topology)

- **Loading**: P = 0.01 MPa → x-compression −15
- **Purpose**: check reproducibility across 3 different mesh realizations at fixed loading

SIM_100 — Test mesh, single run

- **Mesh**: `A00A_Test_mesh.hex2`
- **Loading**: P = 0.01 MPa → x-compression −15
- **Purpose**: validation run on a dedicated test mesh geometry

SIM_1000 — Pressure sweep, 4 mesh realizations, no periodicity

- **Meshes**: `A010` – `A013` (4 realizations)
- **E** = 20 MPa; **P** ∈ {0, 0.005, 0.01, 0.02, 0.04, 0.08, 0.16, 0.32, 0.64} MPa
- **Purpose**: systematic pressure-effect study on a first "medium-size" mesh family without boundary periodicity

SIM_1100 — Pressure sweep, 4 larger mesh realizations, no periodicity

- **Meshes**: `A020` – `A023` (4 realizations, different geometry from A010 family)
- **E** = 20 MPa; same 9-pressure sweep as SIM_1000
- **Purpose**: repeat SIM_1000 study on a second mesh family to check geometry dependence

SIM_1200 — Single mesh, pressure sweep, no periodicity

- **Mesh**: `A020` only (single realization)
- **E** = 20 MPa; 9-pressure sweep
- **Purpose**: isolate pressure effect on one mesh without realization scatter; reference for SIM_1100

SIM_1210 — Single mesh, pressure sweep, no periodicity (created by SIM_9999 script)

- **Mesh**: `A020` (same as SIM_1200)
- **E** = 20 MPa; 9-pressure sweep
- **Purpose**: likely a re-run or variant of SIM_1200; note the creator script is named `SIM_9999_inp_creator.py` but internally starts at sim_num = 1210

SIM_1300 — Pressure sweep, left-right periodic BCs

- **Mesh**: `A030_hexagonal_mesh.hex2`, periodic = `lr`
- **E** = 20 MPa; 9-pressure sweep
- **Purpose**: first left-right periodic series; removes free-edge effects on x-boundaries; enables comparison with SIM_1200 (same loading, no periodic)

SIM_1310 — Pressure sweep, left-right periodic, mesh A030 (second creator)

- **Mesh**: `A030`, periodic = `lr`
- **E** = 20 MPa; 9-pressure sweep
- **Purpose**: separate creator script for A030 l-r periodic; likely a re-generation or variant of SIM_1300

SIM_1320 — Pressure sweep, left-right periodic, mesh A031

- **Mesh**: `A031_hexagonal_mesh.hex2`, periodic = `lr`
- **E** = 20 MPa; 9-pressure sweep
- **Purpose**: left-right periodic study on a second mesh realization (`A031`) for statistical comparison with SIM_1310

SIM_2000 — Fully periodic, single mesh, pressure sweep

- **Mesh**: `A2000_hexagonal_mesh.hex2`, periodic = `both`
- **E** = 20 MPa; 9-pressure sweep; compression via periodic DOF (BC_9999997)
- **Purpose**: first fully periodic series; eliminates all free-edge effects; enables proper bulk-modulus / homogenization analysis

SIM_3000–3228 — Main hexagonal parametric study (primary hexagonal dataset)

- **Meshes**: `A3000`, `A3100`, `A3200` (3 hexagonal geometries, periodic = `both`)
- **E** ∈ {20, 10, 5} MPa; **P** ∈ {0, 0.005, 0.01, 0.02, 0.04, 0.08, 0.16, 0.32, 0.64, 1.28} MPa
- **Total**: 3 meshes × 3 E × 10 P = **90 simulations**
- **Purpose**: primary hexagonal-foam dataset; full sweep of stiffness and inflation pressure across 3 mesh topologies; intended for force-chain analysis, homogenization, and comparison with random meshes

SIM_4000 — Numerical step-size sensitivity study

- **Mesh**: `A3000`, periodic = `both`; E = 20 MPa, P = 0.64 MPa (single condition)
- **Variants**: max increment size ∈ {0.001, 0.0001, 0.00001} → 3 simulations
- **Purpose**: verify numerical convergence; check sensitivity of results to Abaqus time-stepping

SIM_5000–5428 — First random-mesh dataset (primary random dataset 1)

- **Meshes**: `R6000`–`R6004` (5 **random** foam mesh realizations, periodic = `both`)
- **E** ∈ {20, 10, 5} MPa; **P** ∈ {0, 0.005, 0.01, 0.02, 0.04, 0.08, 0.16, 0.32, 0.64} MPa
- **Total**: 5 meshes × 3 E × 9 P = **135 simulations** (present in `E001_Simulations/`)
- **Purpose**: repeat the SIM_3000 parametric study on random (disordered) foam structures; study effect of topological disorder

SIM_6000–6428 — Second random-mesh dataset (primary random dataset 2)

- **Meshes**: `R6010`–`R6014` (5 **different** random mesh realizations, periodic = `both`)
- **E** ∈ {20, 10, 5} MPa; same 9-pressure sweep
- **Total**: 5 meshes × 3 E × 9 P = **135 simulations** (present in `E001_Simulations/`)
- **Purpose**: increase statistical sample of random meshes beyond SIM_5000; validate trends and measure realization-to-realization variability

SIM_9000 — No-pressure test, constrained y-faces, no periodicity

- **Mesh**: `A020`, E = 20 MPa, P = 0 (no inflation)
- **Step-1 BCs**: x-negative fixed in y, x-positive displaced −15 in x, both y-faces pinned in x
- **Purpose**: special boundary-condition test with no gas; explores constrained compression (prevents lateral expansion); differs from all other series in Step-1 BCs

SIM_9001–9005 — Individual test / debug runs

SIM	Mesh	Periodic	P (MPa)	Notes
9001	`TEST_P`	lr	0	Test left-right periodicity on dedicated test mesh, no inflation
9002	`A031`	lr	0.02	Single test on A031 with l-r periodic
9003	`A00B_Test_mesh_rectangle`	ul	0.02	Test on rectangular test mesh with up-left periodicity
9004	`A031`	both	0.02	Same mesh as 9002 but full (both-direction) periodicity
9005	`A991`	both	0.02	Test on mesh A991, fully periodic

Purpose: one-off tests during development of periodic boundary condition implementation.

SIM_9998 — Single mesh A9998, fully periodic

- **Mesh**: `A9998_hexagonal_mesh.hex2`, periodic = `both`
- **E** = 20 MPa, P = 0.005 MPa
- **Purpose**: isolated test on a dedicated mesh; likely used to validate a specific geometry or check a new feature

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Assessment: Which Series Are Useful

Series	Usefulness	Reason
SIM_000–100	Low — prototype	Superseded by larger studies; useful only as minimal reproducible examples
SIM_1000–1320	Low — never ran	Creator scripts exist but no simulation folders; hexagonal non-periodic series was abandoned in favour of periodic approach
SIM_2000	Low — superseded	Single-mesh fully periodic; replaced by larger SIM_3000 sweep
SIM_3000–3228	**High — main hex dataset**	Full E × P × mesh parametric study; never ran but intended as the hexagonal reference
SIM_4000	Medium — numerical study	Useful for justifying step-size choices in SIM_3000/5000/6000
SIM_5000–5428	**High — main random dataset 1**	Ran successfully; 135 sims across 5 random meshes and full E/P range

Series	Usefulness	Reason
SIM_6000–6428	**High — main random dataset 2**	Ran successfully; 135 sims across a second set of 5 random meshes
SIM_9000–9005, 9998	Low — debug/test	One-off tests; keep as reference for boundary condition variants